MODULE 2 - REQUIREMENT ANALYSIS

Step 1: Functional Requirements

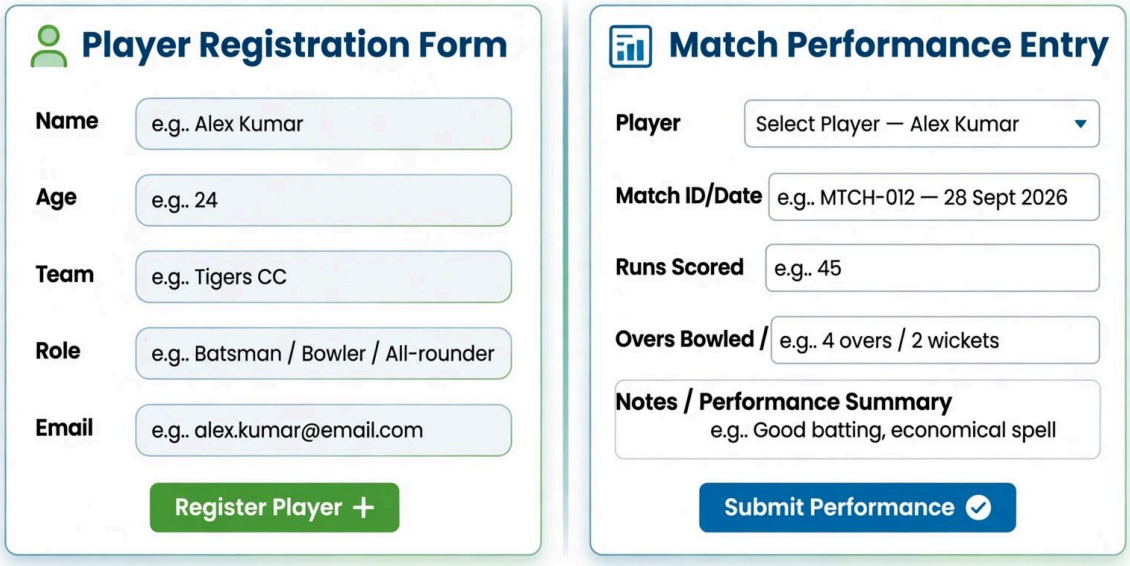
Step 1: Functional Requirements

- **User Registration & Authentication**
  Secure user signup/login, password reset, and role-based access control (Admin, User, Manager)
- **Match Management**
  Create, schedule, and update cricket matches including teams, venue, date/time, and match format (T20, ODI, Test)
- **Live Scoring & Overs Tracking**
  Real-time capture of runs, wickets, overs, extras, and bowler/striker rotation during live match
- **Team & Player Management**
  Add and manage teams and player profiles, track player statistics, roles, and availability
- **Reporting & Analytics**
  Generate match reports, performance summaries, and export data to CSV/PDF for analysis
- **Notifications & Alerts**
  Automated alerts for match start, score updates, results, and system notifications via email or app

MODULE 2 - REQUIREMENT ANALYSIS

Step 2: Player Registration & Performance Entry

Step 2: Define Player Registration & Match Performance Entry



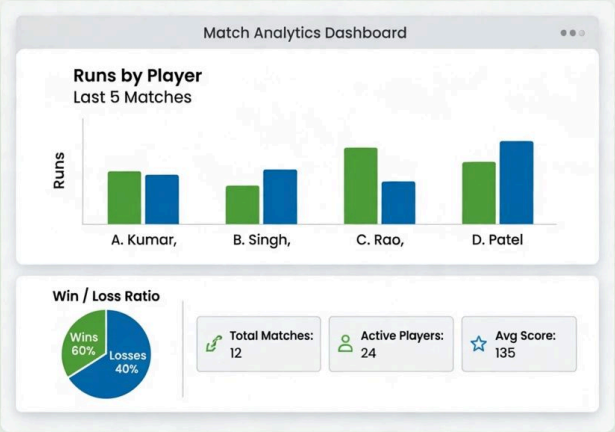
Requirements: Data validated on entry • Secure storage • Linked to player profile & match record

MODULE 2 - REQUIREMENT ANALYSIS

Step 3: Report & Dashboard Requirements

Step 3: Define Report & Dashboard Requirements

Dashboard Mockup



Report Requirements

Match Summary Report
Daily/weekly match report including scores, wickets, key highlights, exportable to PDF/CSV

Player Performance Report
Individual player stats: runs, wickets, average, strike rate, filtered by date/team

Team Comparison Report
Compare team performance metrics across matches, trends over time, visualization charts

Export & Scheduling
Export reports to CSV/PDF/Excel, schedule automated email reports to coaches/admins

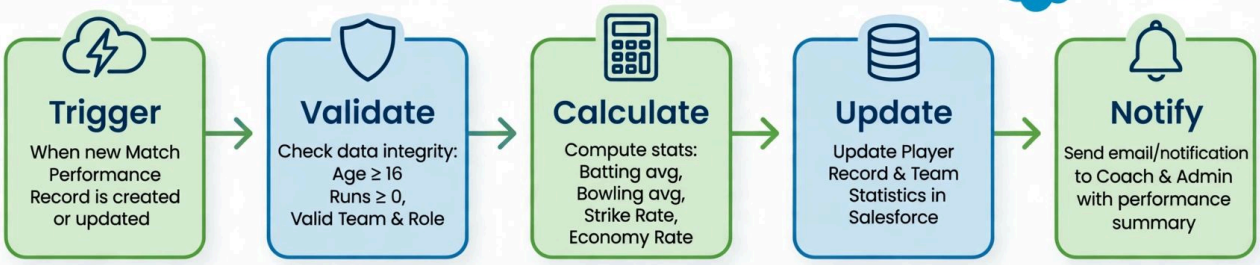
Requirements: Interactive filters • Real-time data refresh • Role-based dashboard access • Responsive design for desktop & mobile

MODULE 2 - REQUIREMENT ANALYSIS

Step 4: Automation with Salesforce Flows

Step 4: Identify Automation Requirements using Salesforce Flows

salesforce



Automation Requirements

- Automate repetitive updates to reduce manual entry
- Ensure validation rules to prevent invalid data before write
- Trigger automated notifications via email/Slack to stakeholders
- Audit trail: log all automated actions for compliance

Requirements: Run in background (asynchronons) • Error handling & retry logic • Role-based notification • Track flow executions in Salesforce Setup

MODULE 2 - REQUIREMENT ANALYSIS

Step 5: Non-Functional Requirements

Step 5: Define Non-Functional Requirements – Security,

Security

- **Data Encryption**
  Secure data at rest and in transit using AES-256/TLS encryption
- **Access Control**
  Role-based permissions, multi-factor authentication for users
- **Audit & Compliance**
  Activity logging, compliance with data privacy regulations (GDPR/Data Protection)

Usability

- **Intuitive UI**
  Clean, responsive design with easy navigation for coaches and administrators
- **Accessibility**
  WCAG 2.1 compliant, keyboard navigation, clear labels and toolips
- **User Support**
  Help toolips, guided workflows, consistent feedback and error messaging

Performance

- **Scalability**
  Handles 1000+ players and 500+ matches with low latency
- **Responsiveness**
  Dashboards load within 2 seconds, real-time updates under 1 second
- **Reliability**
  99.9% uptime, automated backups, monitoring and error recovery

Overall Goal — Ensure the system is secure, easy to use, and performs reliably under load to support players, teams, and administrators